

3.1 General properties of waves

Paper 2 — 0625 and 5054. Mark-carrying lines only.

Definitions

Waves transfer energy without transferring matter.

Frequency: the number of wavelengths that pass a point per unit time. Unit Hz. Also the number of wavefronts / crests / complete oscillations per unit time. $f = 1/T$.

Wavelength: the distance between two consecutive identical points, e.g. consecutive crests.

Amplitude: the maximum distance from the mean position.

Wavefront: a line joining points that are in phase / a line along a crest.

Compression: particles of the medium closer together; higher density and pressure.

Rarefaction: particles further apart; lower density and pressure.

One wavelength of a longitudinal wave is compression to next compression (or rarefaction to next rarefaction).

Compression to the next rarefaction is $\frac{1}{2}\lambda$.

$$v = f\lambda$$

Convert to SI, write the equation, substitute, give the unit.

Frequency from a source: number of oscillations or crests divided by the time. Per minute, divide by 60.

Frequency stays the same when a wave changes medium or depth.

If speed decreases, wavelength decreases. If speed increases, wavelength increases.

If speed is constant and frequency doubles, wavelength halves.

Graphs and drawings

Amplitude is the vertical distance from the mean to a crest. Peak-to-trough is $2A$.

Wavelength is one full cycle along the distance axis.

Points one wavelength apart have the same displacement and the same direction of motion.

Each circle from a point source is a wavefront. The gap between neighbouring circles is λ .

On a height–time graph starting at the mean, the first crest is at $T/4$.

Transverse and longitudinal

Transverse: vibration at right angles to the direction of energy transfer.

Examples: electromagnetic radiation (light, radio, infrared, X-rays), waves on the surface of water, waves on a rope, seismic S-waves.

Longitudinal: vibration parallel to the direction of energy transfer.

Examples: sound, ultrasound, waves in a spring shown as compressions, seismic P-waves.

Ultrasound is a longitudinal wave of higher frequency than audible sound.

A cork on a water wave moves up and down.

A candle flame in front of a loudspeaker vibrates along the line from the loudspeaker.

Reflection

At a plane surface, angle of incidence = angle of reflection.

Frequency, speed and wavelength stay the same. Direction of travel changes.

Echo is reflection of sound. Waves hitting a wall are reflected.

Ripple tank: plane barrier, reflected wavefronts have the same spacing.

Sound-reflection experiment: loudest signal when $i = r$.

Refraction

Refraction is due to a change of speed.

Water waves travel more slowly in shallow water than in deep water.

Deep to shallow: frequency the same, speed down, wavelength down, wavefronts closer together, direction bends towards the normal.

Shallow to deep: speed up, wavelength up, direction bends away from the normal.

Wavefronts stay continuous across the boundary.

Ripple tank: glass sheet on the floor to make a shallow region.

Whole tank made shallower: speed down, wavelength down, frequency the same.

One side of a wavefront reaches the boundary first, slows first, and the wavefront changes direction.

Diffraction

Diffraction is spreading of a wave through a gap or past an edge.

Greatest through a gap when wavelength is similar to the gap size: larger wavelength, smaller gap.

Least when wavelength is much smaller than the gap, or the gap is wider.

Decrease frequency at constant speed → more diffraction. Increase frequency at constant speed → less diffraction.

Wider gap → less curved wavefronts.

At an edge, larger wavelength gives more diffraction.

Harbour mouth, doorway, hearing around a wall or hill: diffraction. Longer wavelength diffracts more.

Drawing through a narrow gap: circular wavefronts of the same wavelength, centred on the gap.

Ripple tank

Wooden bar or dipper driven by a motor produces the waves. Lamp above and screen below show the crests.

Reflection: plane barrier. Refraction: glass plate on the floor. Diffraction: gap between barriers, or a single edge.

Energy transfer: a floating object oscillates and does not travel with the wave.

To increase frequency: move the bar up and down faster. Wavelength then decreases.

5054 written marks

State the definition in the syllabus words.

$v = f\lambda$, SI substitution, unit.

Transverse vs longitudinal: state both directions of vibration relative to energy transfer.

Wavefront drawing: continuous fronts, correct spacing, enough fronts.

Ripple tank: how the wave is made and how it is seen.